



Clicks to Fixers — Project Documentation

AI Concept Chatbot | CAPACITI Tech Career Accelerator

Group Members

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GROUP PROJECT

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This document describes the design, implementation, and features of **Clicks to Fixers** — an interactive AI concept learning assistant built for the CAPACITI Tech Career Accelerator practical project. The chatbot was developed as a fully custom web application and covers all required topic areas from **Modules 1–4** of the CAPACITI bootcamp.

1. IMPLEMENTATION APPROACH & PLATFORM SELECTION

**Platform Selected: Custom HTML / CSS / JavaScript (Vanilla Web Application)**

Rather than using a third-party no-code platform such as Dialogflow or Botpress, this project was built as a fully custom, self-contained web application using standard web technologies — HTML5, CSS3, and vanilla JavaScript.

**Full UI Control**

Custom design enables rich multimedia — inline diagrams, quiz cards, colour-coded responses — that no-code platforms restrict.

**Zero Cost**

No API fees, rate limits, or subscriptions. Runs entirely client-side and can be hosted free on GitHub Pages.

**Instant Deployment**

A single HTML file deployed on any static host and accessed via a public URL immediately.

**Technical Depth**

Building from scratch is more representative of the group's data engineering background than configuring a drag-and-drop tool.

**Offline Capable**

The chatbot functions without internet once loaded — accessible in low-bandwidth South African environments.

**Intent Matching**

Multi-pass keyword and word-overlap scoring engine — functionally equivalent to Dialogflow intent classifiers.

2. IMPLEMENTED Q&A PAIRS (20 TOTAL)

#	TOPIC AREA	Q&A TITLE	MODULE
1	Core Terminology	What is Artificial Intelligence?	Module 1
2	Core Terminology	What is Machine Learning?	Module 2
3	Core Terminology	What is Deep Learning?	Module 3
4	AI vs ML vs DL	AI vs ML vs Deep Learning Hierarchy	Module 1
5	Core Terminology	What is Natural Language Processing (NLP)?	Module 3
6	Real-World Applications	Real-World NLP Applications	Module 3
7	Core Terminology	Key Challenges in NLP	Module 3
8	Ethics	AI Ethics Overview	Module 4
9	Ethics	Bias in AI	Module 4
10	Ethics	Privacy Considerations in AI (incl. POPIA)	Module 4
11	Core Terminology	What is a Neural Network?	Module 3
12	Core Terminology	What are Large Language Models (LLMs)?	Module 3
13	Core Terminology	What is Computer Vision?	Module 3
14	Real-World Applications	Real-World AI Applications	Module 1
15	Core Terminology	Overfitting and Underfitting	Module 2
16	Core Terminology	What is Reinforcement Learning?	Module 2

#	TOPIC AREA	Q&A TITLE	MODULE
17	Core Terminology	What are GANs?	Module 3
18	Real-World Applications	AI in Healthcare	Module 1
19	Real-World Applications	AI in Finance	Module 1
20	Core Terminology	What is Supervised Learning?	Module 2

3. CONVERSATION FLOWS

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Flow 1 — Learn about NLP

✓ Required

NLP Definition → Real-World NLP Applications → Key Challenges in NLP
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Flow 2 — Explore AI Ethics

✓ Required

AI Ethics Overview → Bias in AI → Privacy Considerations in AI
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Flow 3 — AI vs ML vs Deep Learning

Bonus

What is AI? → What is Machine Learning? → AI vs ML vs DL Hierarchy
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Flow 4 — Neural Networks Deep Dive

Bonus

What is a Neural Network? → What is Deep Learning? → What are LLMs?
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Flow 5 — Computer Vision Path

Bonus

What is Computer Vision? → What is Deep Learning? → Real-World AI Applications

Each flow is triggered from the left sidebar. After Step 1 loads, chip buttons for Step 2 and Step 3 appear automatically above the input bar, guiding the user through the complete path without re-typing.

4. TECHNICAL SPECIFICATIONS

✓ Follow-up Questions	Every bot response includes "Related" concept buttons that chain to further Q&As on the same topic — supporting 3+ follow-up questions per topic.
✓ Further Learning	Every response includes a green "Further Learning" box recommending the specific CAPACITI bootcamp module to study next.
✓ Module Citations	Each of the 20 Q&A pairs is tagged to a specific Module (1–4) and cited in the response footer with the full module name.
✓ Cross-Linking	"Related" buttons on every response connect concepts — e.g., Neural Networks links to Deep Learning, Backpropagation, and Activation Functions.
✓ Error Handling	Unrecognised queries return a friendly "I didn't quite catch that" message with 5 topic suggestion buttons and sidebar guidance.
✓ Feedback Mechanism	Every bot response includes 👍 Helpful / 🙋 Not Helpful buttons. On click, a confirmation message replaces the buttons.

5. MULTIMEDIA ELEMENTS

- 1 AI / ML / Deep Learning Hierarchy

Nested node diagram showing AI ⊃ ML ⊃ Deep Learning with colour-coded levels and explanatory notes.
- 2 Neural Network Architecture

Visual showing input → hidden → output layers with colour-coded nodes and role labels.
- 3 NLP Processing Pipeline

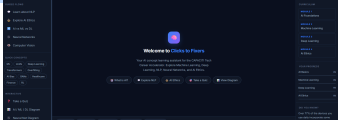
6-step annotated flow from raw text input through tokenisation, preprocessing, to output.
- 4 AI Ethics Framework

6-pillar grid covering Fairness, Transparency, Privacy, Safety, Accountability, and Inclusivity.
- 5 Interactive Quiz

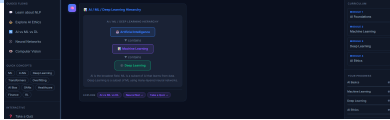
8-question randomised multiple-choice quiz with instant correct/wrong visual feedback and a full explanation per answer.
- All diagrams are rendered inline in the chat window using HTML/CSS — no static images or external files required.

6. SCREENSHOT EXAMPLES

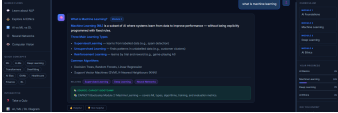
Screenshot 1 — Welcome Screen



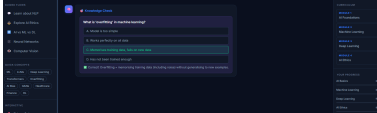
Screenshot 3 — Diagram in Chat



Screenshot 2 — Bot Response with Further Learning

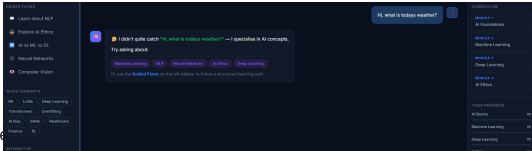


Screenshot 4 — Quiz with Answer Feedback



Screenshot 5 — Error Handling Response

Fallback response for unrecognised question



7. KNOWLEDGE BASE COVERAGE SUMMARY

TOPIC AREA	PAIRS	STATUS
Core terminology (ML, NLP, LLMs, Neural Networks, CV)	11	✔ Exceeds requirement
AI vs ML vs Deep Learning distinctions	2	✔ Met
Real-world applications across industries	4	✔ Met
Basic ethical considerations	3	✔ Met
Total Q&A Pairs	20	✔ Exceeds minimum of 15

8. EVALUATION CRITERIA ALIGNMENT

CRITERION	WEIGHT	HOW THIS PROJECT ADDRESSES IT
Accuracy of AI concept explanations	30%	20 Q&A pairs written to CAPACITI Module standards with structured explanations and key term highlighting
Conversational flow naturalness	20%	5 guided flows with chip-based step progression; related concept buttons maintain natural conversation
Handling of edge case questions	15%	Multi-pass intent matching engine handles rephrased questions; friendly error response with suggestions
Implementation of feedback mechanism	10%	👍 / 🗨️ buttons on every bot response with confirmation message on click
Quality of technical documentation	15%	This 3-page document covering all required sections with tables, flows, screenshots, and coverage summary
Multimedia integration	10%	4 inline diagrams + 1 interactive quiz = 5 multimedia elements (minimum was 3 diagrams + 1 interactive)